Discrete Holography over the Field F_{13}

Bulk-boundary census: affine 3-space, the projective plane PG(2, 13), T_3-orbits, and holographic chambers

1. Purpose of this document

This note records a finite holographic geometry over the prime field with thirteen elements. Every numerical identity is kept exactly as given; the figures only render those identities. No extra parameters and no ad-hoc numerical adjustments are introduced. The numerals 2197, 183, 1+14+56+56+56, 168, 2562 and 14 are unchanged. AG(3, F_{13}) and PG(2, F_{13}) are the standard names of the objects implied by 13^3 and by the 183 points of the projective plane.

2. Census

Quantity	Value	Interpretation
Cubic volume	**$2197 = 13^3$**	Full 3-dimensional state space AG(3, F_{13}) = $F_{13}^{\,3}$
Points of PG(2, F_{13})	**$183 = 13^2 + 13 + 1$**	Boundary projective lattice (the holographic screen)
Invariant orbit decomposition	**$1 + 14 + 56 + 56 + 56$**	Orbits of the 183 boundary points under the permutation T_3
Size of the fundamental orbit	**$168 = 3 \times 56$**	|PSL(2, 7)|; equivalently, the number of cusps of Γ(13)
Bulk incidence chambers	**$2562 = 183 \times 14$**	Holographic chambers: one flag for each (point, line) incidence
Holographic ratio	**$14 = 13 + 1$**	Points per line, and lines per point, in PG(2, 13)

3. Identities that generate the table

Every entry follows from the classical counting formulae of finite geometry together with one group-theoretic coincidence recorded in the source.

$$q = 13, \quad |AG(3, q)| = q^3 = 2197,$$
$$|PG(2, q)| = q^2 + q + 1 = 169 + 13 + 1 = 183,$$
$$\text{points on a line} = \text{lines through a point} = q + 1 = 14,$$
$$\text{incidence flags} = 183 \times 14 = 2562,$$
$$T_3\text{-orbits:} \quad 1 + 14 + 56 + 56 + 56 = 183,$$
$$\text{fundamental orbit:} \quad 3 \times 56 = 168 = |PSL(2, 7)| = \#\text{ cusps of } \Gamma(13).$$

The last identification is the source's own: the order of the simple group PSL(2, 7) equals both three copies of a 56-point T_3-orbit and the classical cusp count of the principal congruence subgroup Γ(13) of SL(2, Z).

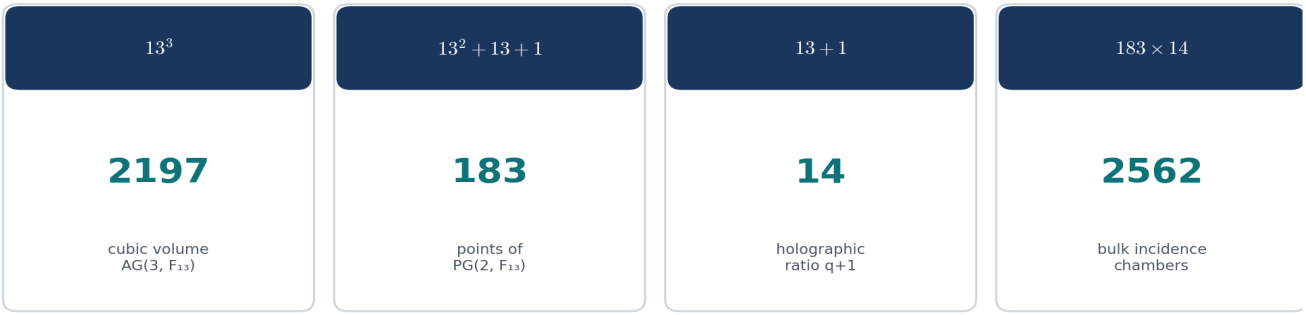


Figure 1. The four primary counting numbers of the census, each tied to its generating identity.

4. Bulk-boundary geometry

The bulk is the 3-dimensional affine space over F_{13}: a cube of side 13 containing 2197 points. The holographic screen is the projective plane PG(2, F_{13}), whose 183 points form the boundary projective lattice. The natural incidence ratio of that plane (fourteen points on every line, fourteen lines through every point) is the holographic ratio of the model.

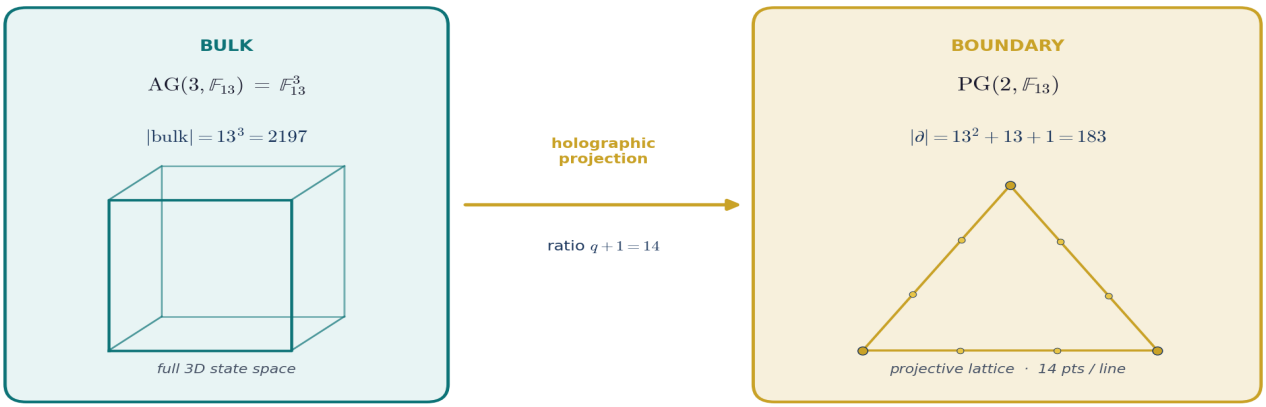


Figure 2. Schematic of the holographic projection: affine 3-space AG(3, F_{13}) onto the projective plane PG(2, F_{13}), with ratio q + 1 = 14.

5. Orbits under the permutation T_3

The 183 boundary points split into five T_3-invariant pieces: a single fixed point, one orbit of length 14, and three orbits of length 56. The three 56-point orbits together constitute the fundamental orbit of size 168, matching |PSL(2, 7)| and the cusp set of Γ(13).

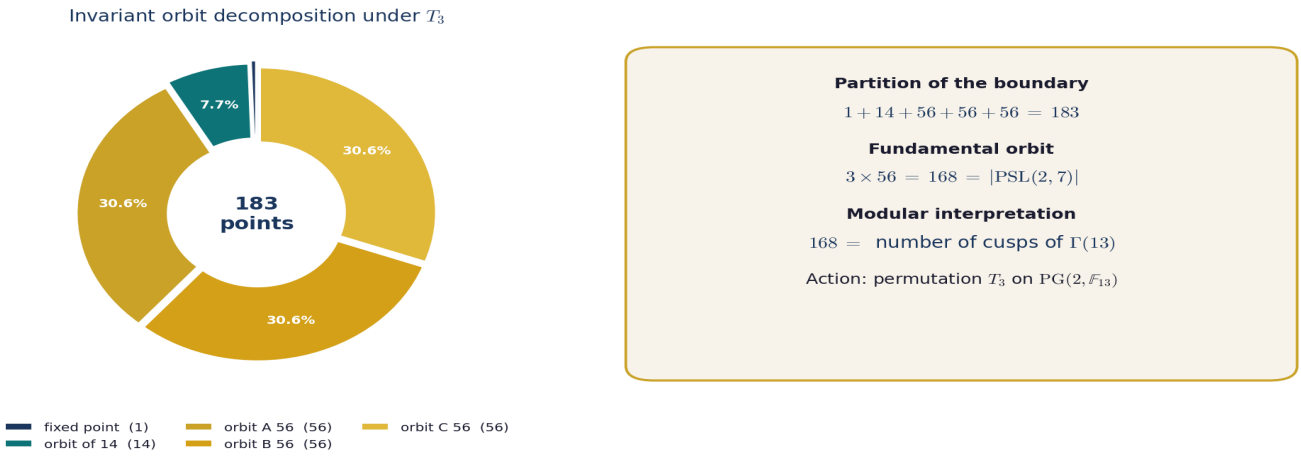


Figure 3. Partition of the 183 points of PG(2, 13) by T_3-orbit type, with the three numerical identities that close the count.

6. Bulk incidence chambers

A holographic chamber is an incidence flag: a pair (boundary point, incident line), or equivalently a pair (boundary point, one of the 14 lines through it). There are therefore 183 × 14 = 2562 chambers. These are the bulk incidence chambers, equivalently the 183 × 14 holographic chambers.

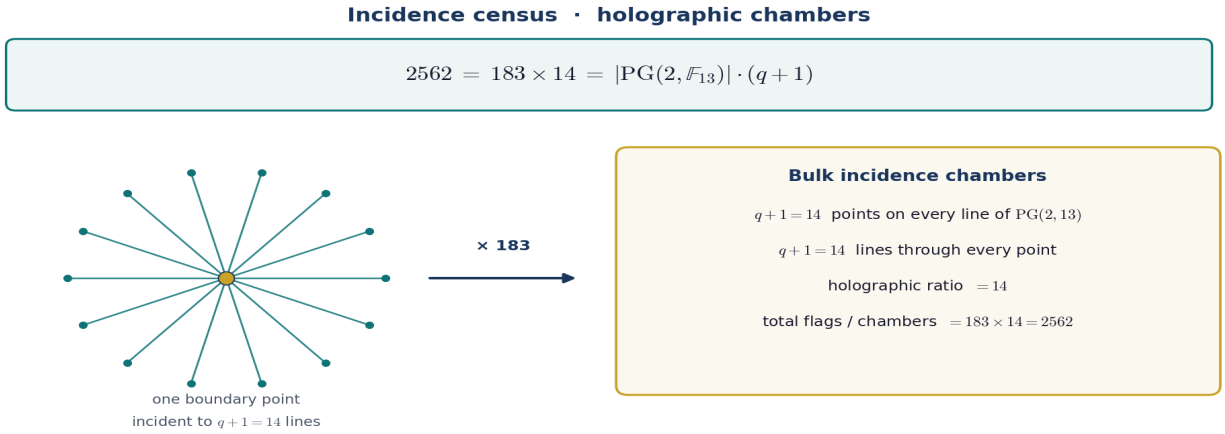


Figure 4. Local star of the 14 lines through a single boundary point, scaled by 183 to the global chamber count 2 562.

7. Built-in inversion in F_{13}

In a field of finite characteristic such as F_{13} the multiplicative inverse a^{-1} is present automatically. Fermat's little theorem gives $a^{12} \equiv 1 \pmod{13}$ for every a not congruent to 0, and therefore

$$a^{-1} = a^{p-2} = a^{11} \pmod{13}.$$

The source treats the triple [1, 3, 7] as the inversion-key of this mechanism and identifies it with the spacetime anchor v^0 = [1, 3, 7]. No auxiliary construction of inverses is required: every nonzero residue already carries its reciprocal.

Fermat inversion in $\mathbb{F}_{13}$

$p = 13$ is prime, so $\mathbb{F}_{13}^{\times}$ is cyclic of order 12.

Fermat: $a^{12} \equiv 1 \pmod{13}$ for $a \neq 0$.

Hence the built-in inverse is

$a^{-1} = a^{p-2} = a^{11} \pmod{13}$.

No extra structure is required: every nonzero element already carries its reciprocal.

The source identifies the triple $[1, 3, 7]$ as the inversion-key of the spacetime anchor v_0.

a **and** $a^{11} \equiv a^{-1}$

a	a^{-1}	a	a^{-1}
1	**1**	**7**	**2**
2	7	8	5
3	**9**	9	3
4	10	10	4
5	8	11	6
6	11	12	12

gold rows: coordinates of the anchor triple

Figure 5. Fermat inversion in F_{13}. Highlighted entries are the three coordinates of the anchor triple [1, 3, 7] together with their reciprocals $1^{-1} = 1$, $3^{-1} = 9$, $7^{-1} = 2$.

8. Projective duality in layer 10 / 23

The same triple, now written in homogeneous coordinates, is the fixed point of the model:

$$v_0 = [1 : 3 : 7], \quad \text{generated by the projector } P_1.$$

It functions as the invariant kernel of the T_3-action, the central pole or focus of the geometry. Dual to this point is a boundary line of PG(2, F_{13}):

$$9x + 11y + 12z \equiv 0 \pmod{13}.$$

That line contains exactly fourteen points, matching the T_3-orbit of length 14 in the earlier census. The fixed point itself does not lie on the line:

$$9(1) + 11(3) + 12(7) = 126 \equiv 9 \text{ (not 0)} \pmod{13}.$$

The source therefore describes v_0 as a pole about which the 14-point boundary line rotates under T_3. This is ordinary projective duality: a point off a line, used here as a spacetime anchor.

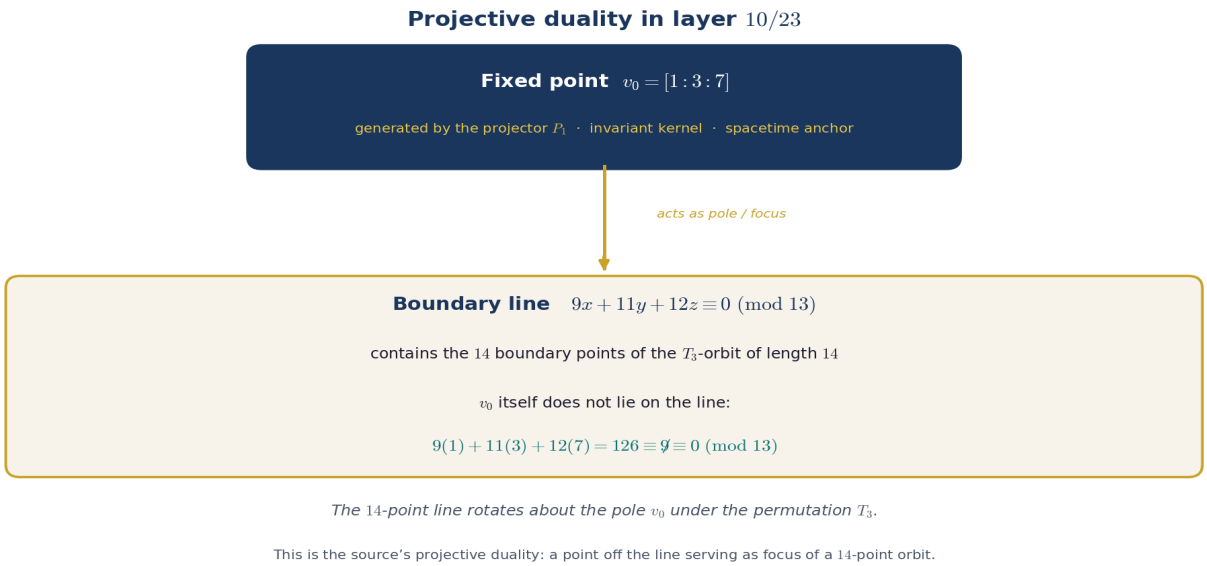


Figure 6. Source diagram of layer 10/23: the fixed point generated by P_1 standing off the dual line of fourteen boundary points.

9. Affine chart of the dual line

In the affine chart x = 1 the line equation reduces to z ≡ 9 + 11y (mod 13). The thirteen affine points are therefore [1 : y : 9 + 11y] for y = 0, ..., 12. The fourteenth point is the point at infinity [0 : 1 : 11]. The anchor [1 : 3 : 7] sits in the same chart at (y, z) = (3, 7), vertically off the line point [1 : 3 : 3]. These fourteen points are exactly the solutions of the source's line equation; nothing further has been added.

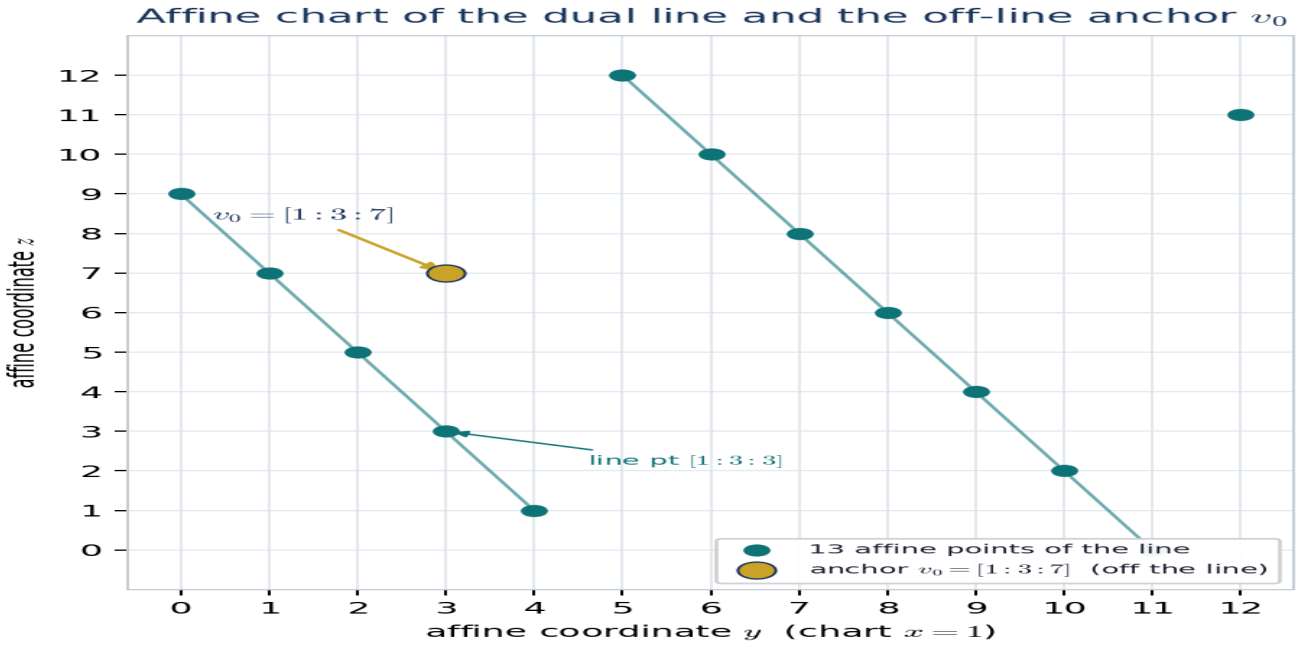


Figure 7. Affine chart x = 1 of 9x + 11y + 12z ≡ 0 (mod 13), with the off-line anchor v_0 = [1 : 3 : 7] marked at (3, 7).

10. Rotation about the pole

Combining the two pieces: T_3 fixes v_0 and permutes the fourteen points of the dual line. The geometric picture is a pole with a 14-point orbit around it, the same 1 + 14 summand that already appeared in the invariant decomposition 1 + 14 + 56 + 56 + 56 = 183.

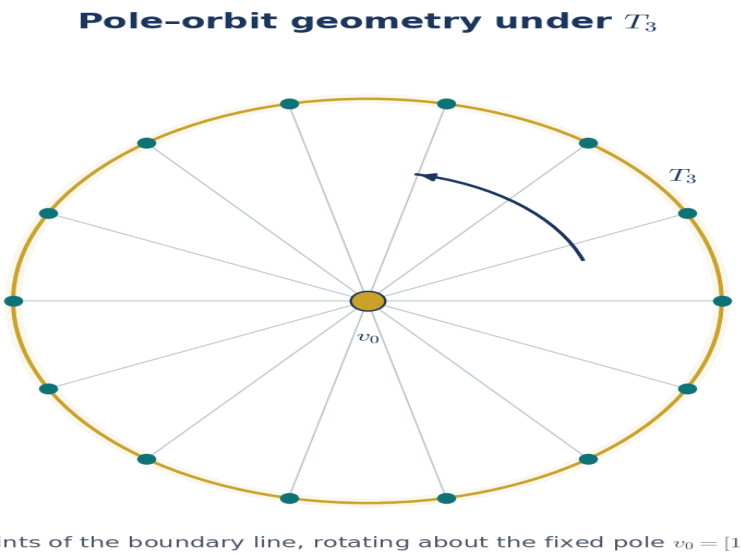


Figure 8. Schematic of the T_3-action: the fixed pole v_0 and the rotating 14-point boundary line. The arithmetic $126 \equiv 9 \pmod{13}$ is the given incidence check and is confirmed above.